

LAB ASSIGNMENT III
NATURAL LANGUAGE PROCESSING LAB (PMDS606P)
CLASS ID - VL2025260105170

MAXIMUM MARKS: 10

DUE DATE: 12 SEPTEMBER 2025

Problem Statement:

Define a function called **processURL** which takes a parameter **URL** which is a valid URL link to a public static web page. Perform the following tasks –

1. Read the text content from the given link and store the content in the variable **textContent**.
2. Preprocess the data in **textContent** and keep only valid English dictionary words.
3. Tokenize all the words and store them in **tokenizedWords** and convert them into lower case.
4. Find the total number of words in **tokenizedWords** and store the result in **numberOfWords**.
5. Find the total number of unique words in **tokenizedWords** and store the result in **numberOfUniqueWords**. Store all the unique words in **uniqueWords**.
6. Perform Histogram Analysis on **uniqueWords**.
7. Find the 10 most frequent words and store the result in the variable **maxFreq**.
8. Generate a word embedding for the web page using only **uniqueWords** (use 100 dimensions).
9. Visualize the word embeddings in a 2D plane.
10. Find out the few cases of most similar words.